Data-Gathering Plan v3 — dataset-grain worldwide TODO (historical included)

> **Method (this build).** 12 parallel domain-expert agents + 2 verified library censuses (CCXT `exchanges.json`, AKShare docs) → 13-column dataset-grain TODO rows (`| # | dataset | class | source & route | granularity | history depth | latency | integration pattern | license class | quality/PIT gates | wave | TODO action | verification |`), machine-merged, deduped, schema-validated into DATASET-TODO (+ PDF, landscape) and `Research/data/extras/data-catalog.jsonl`. **No dataset downloads** — doc pages only. Row verdicts: VERIFIED-UPSTREAM-DOC **357** · SURVEY-EVIDENCE **295** · UNVERIFIED-CANDIDATE **348** (each row labelled; verified rows carry per-volume URL footnotes). Wave mix: W0 **311** / W1 **294** / W2 **243** / W3 **152**. 1,000 dataset rows across 12 volumes.

1. Goal and scope framing

The catalogue targets a collection surface that unions **every tradeable venue class, every filings jurisdiction, and the non-market dataset classes that single vendors do not combine** (registries, litigation, sanctions, patents, clinical, procurement, power-grid ops, shipping, satellites, own-capture moats) — with **historical/past availability stated on every row** (years + whether the backfill is obtainable now). Where a dataset cannot be obtained today, the row says so and names the wave/gate that unlocks it (`quote`, `license-review`).

2. Integration patterns (how a row lands in the system)

pattern	mechanics	auth / rate budget	landing format
REST bulk / paging	page-walk with checkpoint table (`dataset_key`, last pointer)	API key + per-minute cap tracked per token	parquet staging → L0
WS stream	persistent sockets, reconnect-safe seq gaps	per-conn instrument budget (e.g. Dhan 5×5 000)	append-only event table
S3-archive walk	day-partitioned object listing → ranged GET	keyless or signed	file → parquet + hash
SFTP / file-drop	schedule-based directory diff	cred rotation	file → schema-checked table
BigQuery / table export	read API cron, column-pruned	quota caps	parquet via Arrow
library (CCXT / akshare / yfinance)	versioned wrapper configs, function=endpoint map	shared politeness budget	normalized OHLCV/event frame
own-node / subgraph	chain sync + eth_getLogs / subgraph polling	disk budget, RPC caps	derived tables only (PB-raw never stored)
own-capture	cron fetcher + fetch-ts + raw-blob store	politeness + robots respect	treasure L0 with Two-Clicks lineage
licensed terminal / feed	vendor SDK, per-seat	seat + redistribution contract	internal-only landing (no-local-store respected)
STAC / object search	catalog search → item fetch	public	GeoParquet/COG

Storage spine stays vol-10's `L0 VAULT → L1 REFINERY → L2a/b/c BRAIN → L3 FACES`: append-only parquet + hash + versioned, golden-file tested marts, factors/events/edges leaves, one face.

3. Cross-cutting hard gates

1. **Two-Clicks lineage** — every table keeps source-URL + fetch-ts + parser-version columns.
2. **Honest freshness** — declared latency + upstream publication lag; freshness class per row.
3. **License is a build gate** — free / internal-only / commercial-redistribution-ok / no-local-store / vendor-license / □-blocked-never-commercial. `no-local-store` rows never persist as files; NSE scraping rows are □-blocked-never-commercial (display path = TrueData/GlobalDatafeeds-class W1 rows).
4. **PIT / survivorship** — □NEVER-PIT rows (FMP/EODHD/Tiingo/SimFin overwrite-in-place) never feed the backtest factor spine; survivorship-correct universes (norgate/CRSP-class) gate any factor backtest.
5. **Same skeleton** — 13-column row contract everywhere; new sources enter as rows, not code.
6. **QA in the merge** — schema check (13 cols), integer #, wave/action/verification allow-lists, garble scan, dedupe on canonical dataset key (dupes dropped with build-log note).

4. Backfill order (what unblocks what)

order	domain	sizes (owner-claim sizing)	unblocks
1	benchmark bundle < 5 GB (Pink Sheet, EIA, IMF/WB, FAO, OWID, WHO)	instant	cross-edge joins
2	India eq + F&O bhavcopy archives (1994/2000→)	~150 GB	India depth product
3	crypto tick crown-jewel (binance.vision top-300 ≈ 600 GB; on-chain derived)	L0-alpha	microstructure research
4	FX tick (Dukascopy 2003→) + fixes	~50 GB	intraday FX
5	GDELT retro news spine (1979→) + day-1 self-archive RSS rails	index-scale	event factors
6	filings (EDGAR full, EDINET, DART XBRL, Companies House, cninfo)	~30 GB	fundamentals PIT
7	power/grid + weather grids	streaming+history	infra factors
8	registries/legal/IP/clinical/procurement archives	~10 GB	moat joins (L2c edges)

5. Census instruments (verified this session)

- **CCXT `exchanges.json` — 101 exchange ids with unified REST + 71 with WebSocket** verified live from upstream `exchanges.json` (2026-10-24), plus a native **prediction** interface family covering 9 venues (polymarket, kalshi, limitless, myriad, opinion, predictfun, sxbet-class) — closes most of the prediction-market gap without separate vendor work.

- **AKShare docs** — open Python interface library for stocks/futures/options/funds/FX/bonds/indices/crypto from public sources, per-endpoint docs, HTTP wrapper (AKTools); docs updated 2026-10-23. Academic-research framing — treat redistribution like scraped-source re-syndication (attribute, no commercial resale of raw rows).
- 12 agent volumes with per-row verification labels and ≤30-fetch budgets each.

6. The TODO table

The actionable artefact: DATASET-TODO.md — 1,000 rows, one per dataset, with `TODO action` per row (connect / backfill / verify / quote / license-review / build-parser). PDF: `Research/data/todo/DATASET-TODO.pdf` (landscape). Domain volumes:

volume	domain	rows	volume file
01	India equities & F&O	50	01-*.md
02	Asia-Pacific equities	90	02-*.md
03	US equities & microstructure	81	03-*.md
04	EU & global-rest equities	72	04-*.md
05	Fundamentals · filings · PIT	121	05-*.md
06	Derivatives (options/futures)	84	06-*.md
07	FX · rates · macro · credit · ESG	103	07-*.md
08	Commodities · energy · power · shipping	86	08-*.md
09	Alternative data	75	09-*.md
10	Registries · legal · IP · clinical · procurement	72	10-*.md
11	Crypto full-stack	90	11-*.md
12	Cross-domain historical archives	76	12-*.md

7. Verdicts (honest counts)

- VERIFIED-UPSTREAM-DOC 357 rows (live-fetched pages this session, URLs footnoted per volume).
- SURVEY-EVIDENCE 295 rows (traced to the committed 108-dossier corpus/matrix).
- UNVERIFIED-CANDIDATE 348 rows — candidates with `verify`/`quote` actions, labelled as such.
- Build-log: 0 schema-malformed rows (all 1,000 parse 13/13 columns) schema problems, 1 (FINRA daily short-sale volume — kept in the 03 market-structure volume, dropped from 05) duplicates dropped (transparent, in build log).

8. Annexes

- ALPHALENS-INTAKE — requirements source (6 classes, spine, hard rules)
- DATA-SOURCE-MATRIX / V2 — vendor-grain lineage (273 rows)
- `Research/data/streams/01…06` — the first-pass vendor census feeding V2
- `Research/data/extras/data-catalog.jsonl` — this plan's machine-readable rows